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Printing biological solutions through laser-induced forward transfer

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Abstract Laser-induced forward transfer (LIFT) is a direct-writing technique adequate for the high-resolution printing of a wide range of materials, including biological molecules. In this article, the preparation through LIFT of microarrays of droplets from a solution containing rabbit antibody immunoglobulin G (IgG) is presented. The microarrays were prepared at different laser pulse energy conditions, obtaining microdroplets with a circular and well-defined contour. The transfer process has a double threshold: a minimum energy density required to generate an impulsion on the liquid film, and a minimum pulse energy, which corresponds to the onset for material ejection. In addition, it was demonstrated that the transfer process can be correctly described through a simple model which relates the energy density threshold with the amount of released material. Finally, a fluorescence assay was carried out in which the preservation of the activity of the transferred biomolecules was demonstrated.

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1 Introduction

Direct-writing techniques that use a laser as a writing tool have been successfully applied to print several materials onto a variety of substrates [1]. Lasers offer interesting properties such as high focusing power and the fact that they avoid any mechanical contact. This allows obtaining higher degrees of integration than other more extended direct-writing techniques, such as ink-jet printing [2], with the added advantages of avoiding contamination issues and clogging problems. Among the laser-based techniques, laser-induced forward transfer (LIFT) constitutes an interesting alternative [3, 4].

In LIFT, a laser pulse is used to transfer a portion of a donor film (previously deposited on a laser-transparent holder) onto a receiving substrate, placed parallel and in close proximity to the film. In the case that a liquid film is used [5], the laser pulses can propel the material in a gentle way, avoiding significant phase change and preserving it from decomposition. This makes LIFT suitable for printing biological solutions, which allows the application of the technique to the preparation of miniaturized biosensors.

The studies of the LIFT of liquid films have been mainly aimed to prove the feasibility of this technique to transfer different biological solutions without harm, such as proteins [6–9], DNA [10–12], living cells [13–15] and even microorganisms [16]. Some studies have also been focused on the transfer process, either using CCD time-resolved imaging [7, 17, 18] or through the characterization of the deposited droplets under the systematic variation of different experimental parameters [9, 11, 19]. In this last work [19], we proposed a model which successfully described the transfer process for a biomolecule-free solution.

In this paper we report the preparation of microarrays of droplets of an immunoglobulin G (IgG) solution through LIFT at different laser pulse energies. Immunoglobulin G is the most common antibody, and it has the capability to adapt and recognize different antigens, which makes it useful in many clinical applications, such as medical diagnostic

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or disease therapy. The study of the LIFT of an IgG solution is carried out through the optical characterization of the deposited droplets, which allows testing the reported model in the case of a biomolecule-containing solution. Finally, the biological activity of the transferred proteins is analysed.

2 Experimental

A pulsed Nd:YAG laser tripled in frequency (355-nm wavelength, 10-ns pulse duration, 1-Hz repetition rate) was used to transfer droplets of a 0.05 mg/mL rabbit IgG solution where the solvent was a mixture of PBS (0.01 M, pH 7.4) at 60% (v/v) and glycerol at 40% (v/v). A volume of 20 µL of this solution was spread on top of a holder by means of a blade coater, resulting in a 5-µm-thick liquid film (thickness estimated by measuring the film weight). The holder consisted of a laser-transparent glass slide previously coated by thermal evaporation with a titanium film around 60-nm thick. This film acts as an absorbing layer, needed due to the transparency of both solution and glass slide to the laser radiation. Several materials such as titanium [6, 9–12, 19], gold [7], silver [16] or even polymers [14] have been used as absorbing layers to print different biological solutions without observing loss of bioactivity.

The holder–liquid film system, usually named a ribbon, was then placed parallel to a receiving substrate separated 30 µm from it, with the liquid facing the substrate surface. This substrate consisted of a commercially available poly-L-lysine-coated glass slide, a kind of substrate commonly used for the immobilization of the biomolecules. Both ribbon and substrate were then placed on a computer-controlled translation stage, allowing their precise displacement with respect to the laser beam. Using an in-house computer program, both laser firing and translation of the ribbon–substrate system were synchronized, so that every single laser pulse produced a unique droplet. Microarrays of droplets could be obtained in this way.

The laser beam presented an elliptical Gaussian energy density distribution (F), given by the expression

$$F(x,y) = \frac{2E}{\pi \omega_x \omega_y} e^{-2\left(\frac{x^2}{\omega_x^2}+\frac{y^2}{\omega_y^2}\right)} \quad (1)$$

where E is the laser pulse energy, ω_x and ω_y the major and minor beam radii, respectively, and x and y are the Cartesian coordinates along the major and minor axes of the beam, respectively. The laser beam was focused on the titanium film with a ×15 microscope objective.

In a first experiment, a microarray of 11 rows per 21 columns was prepared varying the laser pulse energy from row to row and from 1 µJ to 8 µJ. The dimensions of the laser beam in this experiment were $\omega_x = 27$ µm and $\omega_y = 19$ µm; they were determined measuring the radii of the spot obtained on the titanium film at different laser pulse energies, and through the linear fit of the radii squared versus the logarithm of the energy. In a second experiment, a 10 rows per 9 columns microarray was obtained using a different beam with dimensions $\omega_x = 47$ µm and $\omega_y = 30$ µm. The energy was also varied from one row to another, with values ranging from 10 to 60 µJ.

Optical microscopy was used to characterize the morphology of the deposited droplets. An optical profiler allowed measurement of their three-dimensional profiles, which were then employed in the calculation of their volumes. A fluorescence test was performed to evaluate the bioactivity of the transferred droplets by using cyanine molecules (Cy5) as fluorescence tags, and detection upon excitation at a wavelength of 635 nm.

3 Results and discussion

3.1 Transfer process

All the deposited droplets in the first microarray present a circular and well-defined contour (Fig. 1). Their shapes, analysed through optical profilometry, correspond to spherical caps. It is also observed that the diameters of the transferred droplets increase with increasing laser pulse energy, as has been found before with other transfer solutions [10, 19]. However, diameters cannot be minimized indefinitely: there exists an energy threshold, $E_{\min}$, below which no material is transferred. It is for energies slightly above this value that the minimum droplet diameters are obtained, and thus the maximum resolution of the technique is achieved for given laser beam dimensions. In this case,

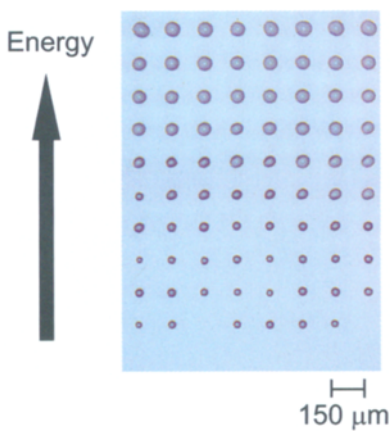


Fig. 1 Optical microscopy image of a microarray obtained through LIFT of an IgG solution prepared varying the laser pulse energy. The *arrow* on the *left* indicates the direction of increasing energies of the rows. The row corresponding to the lowest energy is completely free of droplets

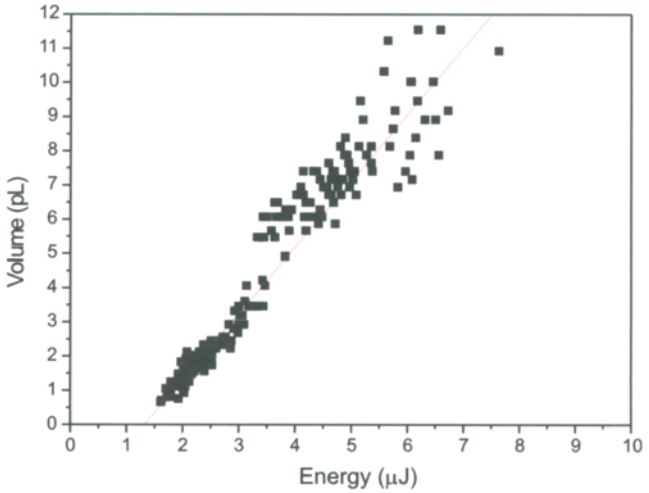


Fig. 2 Plot of the deposited droplet volumes versus the laser pulse energies. The *solid line* corresponds to the linear fit

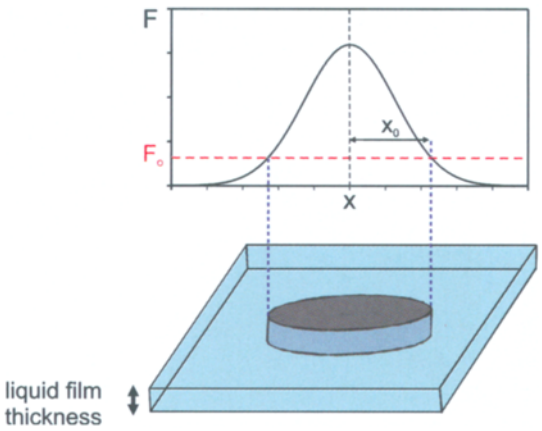


Fig. 3 Schematic representation of the cylinder considered in the model

droplets of 25 μm in diameter were printed, a value similar to that obtained with other biological solutions on identical substrates [9–12, 19], and considerably smaller than the minimum ones obtained with commercial arrayers [7].

In order to quantify these results, the droplet volumes were calculated and plotted versus the laser pulse energy that caused the ejection (Fig. 2). It is observed that the points present an upward trend, with a linear dependence between droplet volumes and pulse energies. Such a linear relationship has also been found for different solutions [9, 19], and linear dependences between the amount of transferred material and different parameters related to the energy delivered to the liquid sample have been described for other printing techniques [20]. However, in the present case two clearly differentiated regimes can be observed: for energies up to 4 μJ all points are well aligned, with little dispersion, while, for higher energies, a significant and sudden increase in the dispersion of the droplet volumes is produced, as well as a slight shift toward upper volumes. This change might be due to the activation of a different transfer mechanism at high energies, as might occur in the transition from simple Ti vaporization to plasma formation. In fact, changes observed in the back-side ablation behaviour of carbon films at very similar laser fluences have been previously attributed to this transition [21].

The aforementioned linear dependence can be characterized through the simple fit

$$V = K(E - E_0), \tag{2}$$

where V is the droplet volume, E is the laser pulse energy and K and E_0 are the fitting parameters. In this experiment the value of K is 1.9 pL/μJ, and that of E_0 is 1.3 μJ. The parameter K corresponds to the efficiency of the transfer process, whereas E_0 can be used to find an energy density threshold F_0 through the expression [19]

$$F_0 = \frac{2E_0}{\pi \omega_x \omega_y}. \tag{3}$$

This parameter, which is independent of the laser focusing conditions, has a value of 0.16 J/cm^2, similar to those obtained for previously reported solutions [18, 19]. The F_0 parameter is interpreted as the minimum energy density that results in an impulsion on the liquid film. However, only when the total laser pulse energy overcomes $E_{\min}$ is such an impulsion energetic enough to lead to material release. Images obtained through CCD time-resolved imaging have revealed the formation of transient material protrusions without liquid ejection during LIFT experiments [17, 18], which demonstrate the existence of such an impulsion. Therefore, the transfer process is indeed activated by a double threshold: F_0 plus $E_{\min}$ [19].

The existence of the parameter F_0 led to a model [19], which provides a simplified description of the transfer process where the volume of the deposited droplets is directly related to F_0. According to this model, the amount of transferred material corresponds to the volume of the cylinder whose base is formed by the points with an energy density $F(x, y) > F_0$, and its height corresponds to the liquid film thickness d (Fig. 3). In order to test the validity of the model in the current experiment, a plot of the droplet volumes versus the volumes of the cylinders was carried out (Fig. 4). The cylinder volumes were calculated as $\pi x_0 y_0 d$, where x_0 and y_0 are respectively the maximum and minimum radii of the intersection of the Gaussian distribution and F_0 (Fig. 3). As is observed, for cylinder volumes up to 4–5 pL all the points lie close and slightly below the unit-slope line, which means that the model describes rather correctly the transfer process in this situation, that is, in the low-energy range. According to this, in the present case the

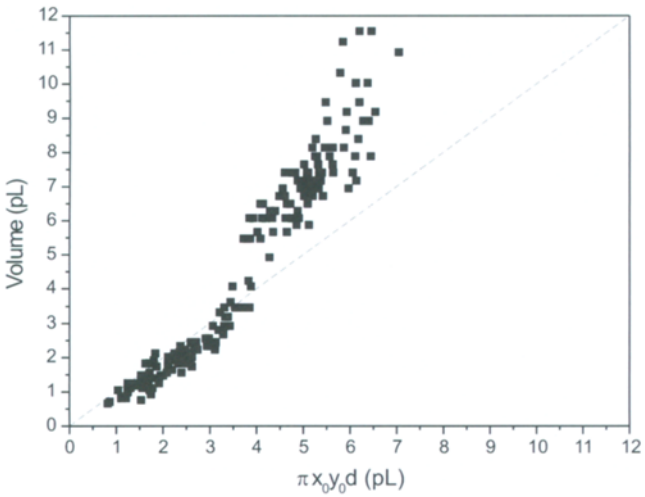


Fig. 4 Plot of the droplet volumes versus the volumes of the cylinders of the model ($\pi x_0 y_0 d$). The *dashed line* represents the unit-slope line

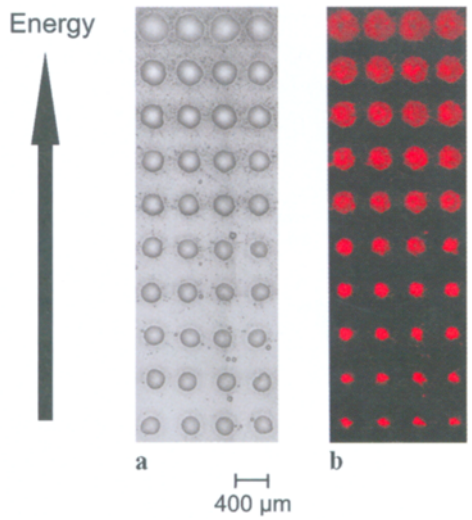


Fig. 5 **a** Optical microscopy image of a microarray obtained through LIFT of an IgG solution. **b** Fluorescence image of the same microarray after application of the fluorescence test with Cy5-conjugated anti-IgG. The *arrow* indicates the direction of increasing energies of the rows

extension of the impulsion generated in the liquid (area with $F(x, y) > F_0$) determines the amount of transferred material. For higher laser pulse energies, however, the volumes of the transferred droplets are shifted above the unit-slope line, which means that the cylinder model cannot be applied in this energy range. This has been attributed [19] to border effects, which become relevant at high laser pulse energies: a significant lateral heat conduction or liquid drag from the borders could be responsible for the generation of an impulsion with effects surpassing the area corresponding to F_0. However, the correspondence between the set of points of the high-energy range in Fig. 2 and those above the unit-slope line in Fig. 4 suggests that the possible activation of the proposed transfer mechanism at high energies could also lead to a higher amount of deposited material than that predicted by the cylinder model, due to the violent expansion of the generated plasma.

3.2 Viability test

A second microarray was prepared in order to test the bioactivity of the transferred proteins (Fig. 5a). This experiment was performed under the same conditions as in the first, but in this case larger laser beam dimensions were used. Therefore, higher energies were needed in order to produce the material ejection. It is observed that droplet diameters increase with the laser pulse energy, and that satellite droplets appear as the energy is raised, which could be due to the increase of the kinetic energy of the transferred material before impact [19].

The slide where the microarray was deposited was immersed after transfer in a 10 mg/mL BSA solution in which the solvent was a mixture of water and glycerol at 50% (v/v). The BSA blocked the poly-L-lysine sites where IgG was not dispensed. The slide was then exposed to a Cy5-labeled rabbit anti-IgG solution solved in PBS at 60% (v/v) and glycerol at 40% (v/v), and submitted to a fluorescence test. The fluorescence image (Fig. 5b) reveals an excellent correspondence between the deposited droplets and the red spots, indicating that the transferred biomolecules bonded to their complementary pairs for all the laser pulse energies and that, therefore, their bioactivity is preserved during the LIFT process.

4 Conclusion

Optical microscopy images of microarrays prepared through LIFT of an IgG solution, containing PBS and glycerol, have showed that spherical-cap-shaped droplets are obtained with a circular and well-defined contour. Quantitative analyses have revealed that the droplet volumes present a linear dependence with the laser pulse energies. There can be distinguished, however, two different regimes: at low laser pulse energies the points are well aligned whereas at higher energies there is a sudden and significant increase in the dispersion and a shift toward upper volumes, attributed to the possible activation of a different transfer mechanism. It has also been found that there exists an energy density threshold, F_0, which has to be overcome to produce an impulsion on the liquid film but, only for pulse energies higher than a certain value, $E_{\min}$, does it result in liquid ejection. Our previously proposed cylinder model has been found to correctly describe the transfer process at the low-energy range, as it did for a biomolecule-free solution. At high energies, however, the amount of transferred liquid is greater than that predicted by the model.

Moreover, the fluorescence test has revealed that the capacity of the deposited proteins to bond to their complementary pairs is preserved and, thus, that the biomolecules' activity is not degraded during the transfer process. Therefore, it can be concluded that indeed LIFT is a viable technique for IgG microarray preparation.

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